

Digital Trainer Circuit

Familiarization of Digital Trainer kit, Logic Pulser, Logic Probe & Digital ICs IE 7400, 7402, 7404, 7408, 7432 & 7486.
(draw their pin diagram and features)

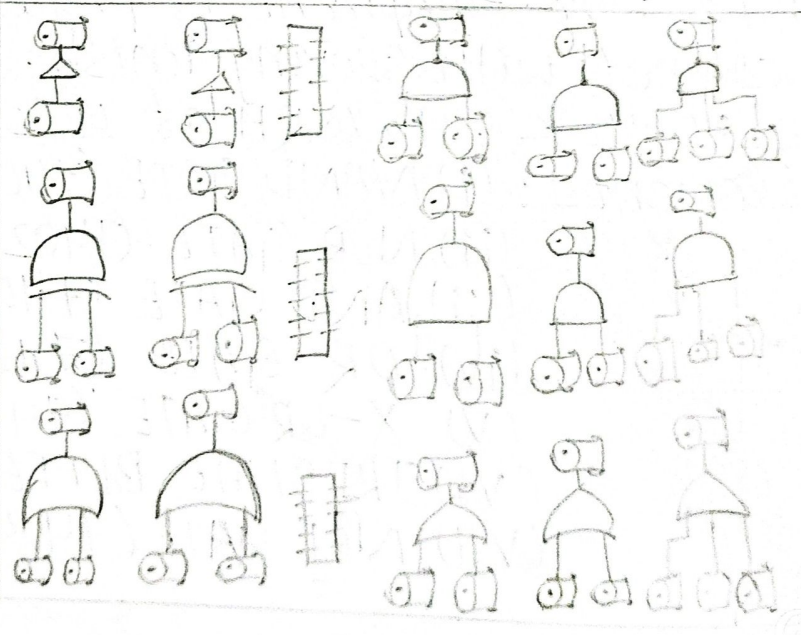
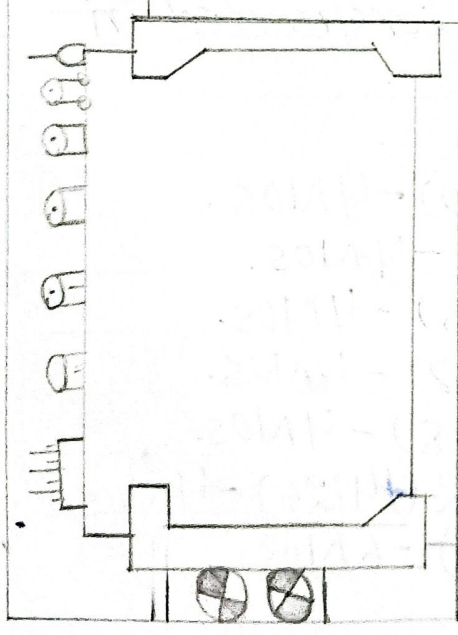
★ Equipments required:-

- (I) Digital Trainer kit
- (II) Power Supply.

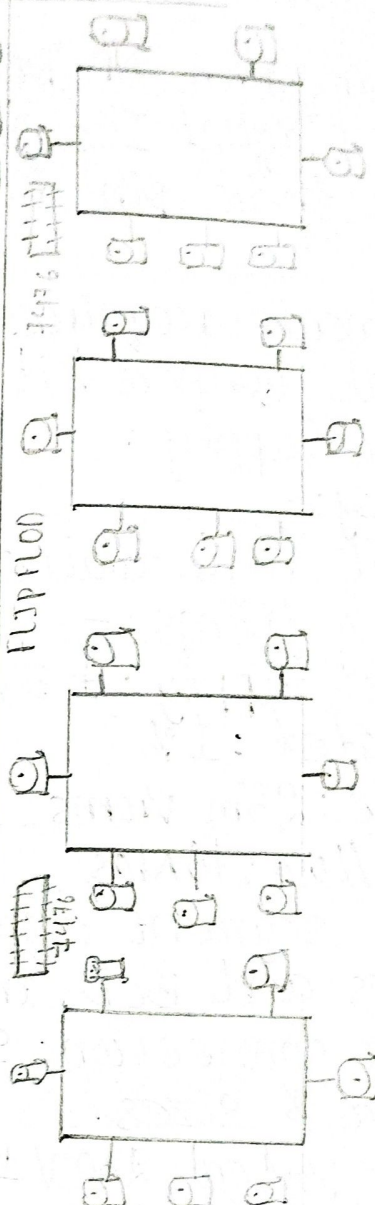
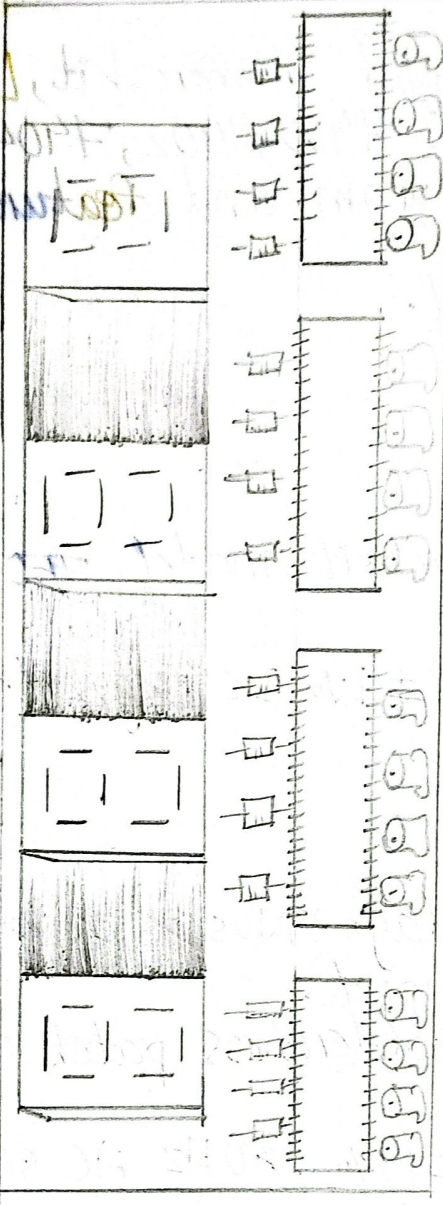
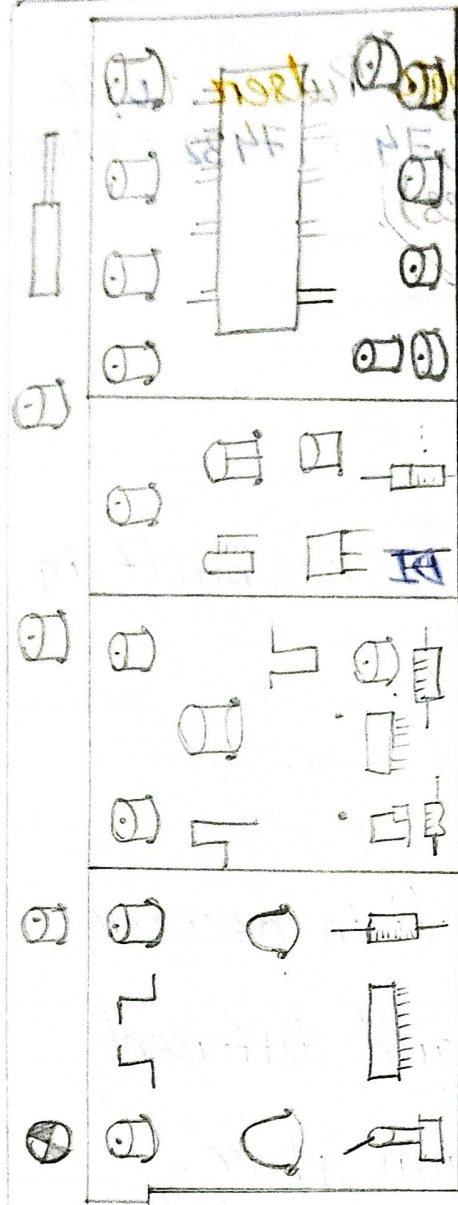
★ Theory:-

Digital logic trainer kit model 750 DI has built in following items:-

- Power Supply: $\pm 5V \pm 1\%$, DC regulated.
- Regulated: 1%
- Ripple: 25m Vrms
- High/Low: 16 Nos
- Seven segment display: 3 Nos. with built in decoder drives and BCD input.
- Inter connection: solderless patch card different colour & sizes.
- Input power: $230V \pm 10\%$, 50 Hz AC signal phase.
- Mechanical: The equipment is phase constructed in metal cabinet with sloving finish.
- Ambient: $0-40^{\circ}\text{C}$, 95% RH at 40°C .
- Logic Elements:-
 - (i) NAND GATE (7400) - 4 Nos.
 - (ii) NOR GATE (7402) - 4 Nos.
 - (iii) AND GATE (7408) - 4 Nos.
 - (iv) OR GATE (7432) - 4 Nos.
 - (v) X-OR GATE (7438) - 4 Nos.
 - (vi) TRISTATE BUFFER (74126) - 4 Nos.
 - (vii) NOT GATE (7404) - 6 Nos.



Micro System
PVT. LTD
DIGITAL IC
LOGIC TRAINER



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- Logic elements: (viii) J.k flipflop (7473) - 2 Nos.
- (ix) Schmitt's Trigger (7413) - 2 Nos.
- (x) Flipflop (7472) - 2 Nos.
- (xi) Expander (7460) - 2 Nos.
- (xii) AOL GATE (7450) - 2 Nos.
- (xiii) Gated J.k flipflop (7470) - 1 Nos.
- (xiv) Binary Adder (7483) - 1 Nos.
- (xv) 12 counter (7492) - 1 Nos.
- (xvi) 16 Counter (7493) - 1 Nos.
- (xvii) BCD counter (7490) - 1 No
- (xviii) Shift register (7493) - 1 No
- (xix) 8-bit DAC - 1 No
- (xx) 8-bit ADC - 1 No
- (xxi) Decoder (74154) - 1 Nos
- (xxii) Encoder (74151) - 1 Nos

All the IC's are mounted on IC bases, soldered, in PCB for easy service and ability it's possible to replace the IC without opening the timer board. The IC is visible from the top and the pin extensions are available around. Provision is there to connect external digital voltmeter to achieve better accuracy of reading.

Conclusion:-

From the above experiment, we have studied about different functions of digital trainer kit.

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Date of Submission:- 22.08.24

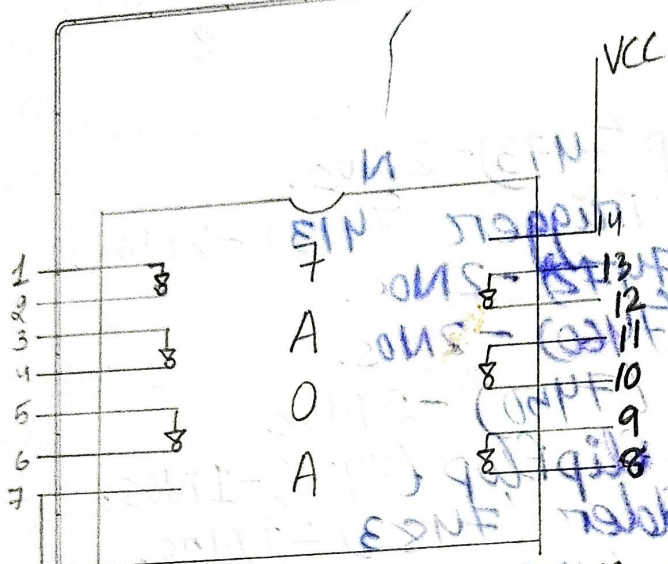
Teacher's Signature

A+

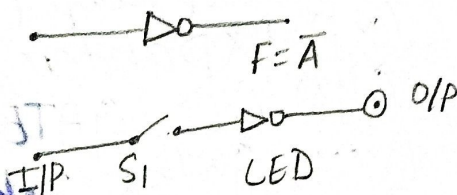
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Galaxy 170

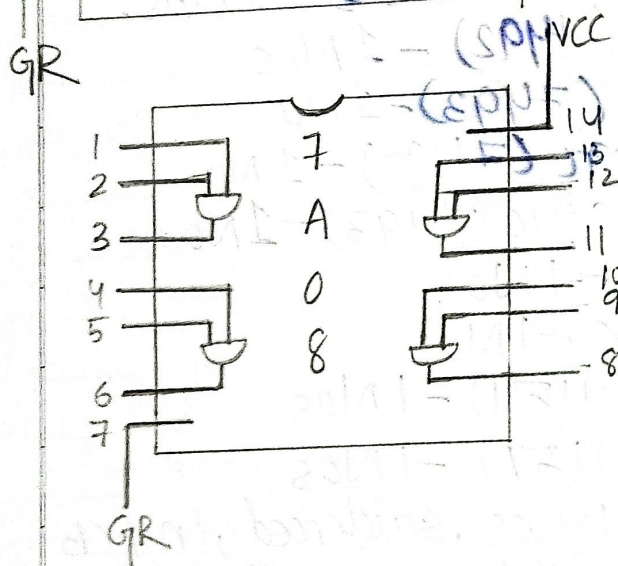
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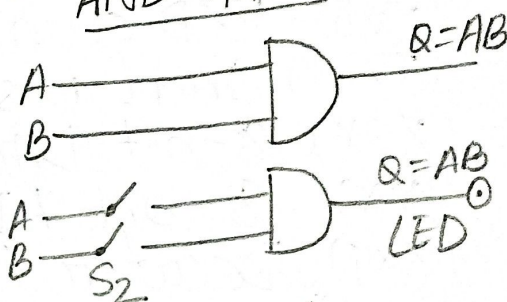
NOT GATE



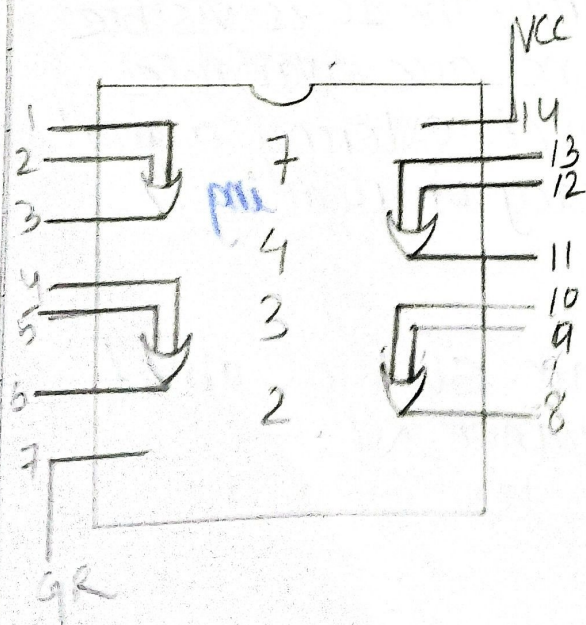
I/P	O/P
A	$\bar{A}$
0	1
1	0



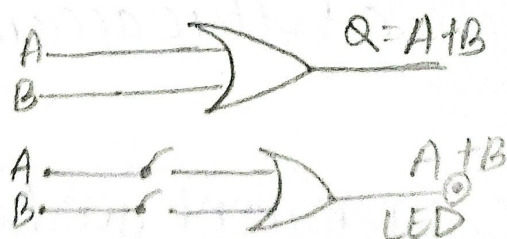
AND GATE



I/P		O/P
A	B	A.B
0	0	0
0	1	0
1	0	0
1	1	1



OR GATE



I/P		O/P
A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1